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In Situ Observations of the Gametophytes of Six Fern Species of Costa Rica

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Abstract.—Fern gametophytes are an often-overlooked part of field studies on ferns due to the challenge of finding material and the difficulty of identifying wild-collected plants. Most existing gametophyte descriptions are derived from lab-reared plants and while it is thought that many of the morphological features are similar under both conditions, we still have a poor understanding of *in situ* morphology for most taxa. The goal of this work was to document a set of morphological observations from the gametophytes of six different species collected *in situ*. We discovered identifiable gametophyte populations of *Elaphoglossum latifolium*, *Hymenophyllum myriocarpoum, Lomariopsis maxonii, Polypodium dulce, Polytaenium cajenense,* and *Sticherus bifidus* during the 2022 OTS Tropical Ferns and Lycophytes course in Costa Rica. Gametophytes were collected in the field, observed, and photographed using stereo and compound microscopy. This work represents the first detailed description for these species' gametophytes from field collections and offers insights into their natural morphology.

Key Words.—gametophyte, morphology, *in situ* collections, tropical, Costa Rica

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The last two decades have seen a significant increase in studies focused on developing a better understanding of fern biology. A growing trend has been the incorporation of gametophyte biology, which has driven a greater appreciation of ferns as complete organisms. Such work has been welcomed as gametophytes have historically received less attention compared to the sporophyte, partly owing to their small size, ephemeral nature in temperate climates, locations where much of pteridology has been focused (*e.g.,* Pelosi and Sessa, 2021), and the general difficulty of identifying field collected material. Recent work has clearly demonstrated the importance of gametophyte field identification in understanding both the ecology (*e.g.,* de Groot *et al.,* 2011; Harrington, Blake-Mahmud, and, Watkins, 2022) and evolution of ferns (*e.g.,* and Moran 2019; Nitta *et al.,* 2021).

Despite growing appreciation of the gametophyte generation, most of our knowledge of fern ecology is still based on observations of sporophytes, which are usually larger and easier to study and identify than gametophytes. However, field and lab work focusing on the ecology of fern gametophytes have clearly demonstrated the feasibility of working with this stage and has generated novel insights into fern biology. This phase of the life cycle is highly variable (reviewed by Farrar *et al.* 2008), with gametophytes occupying a range of growth forms and habits (*e.g.*, subterranean vs. terrestrial, short vs. long-lived) at times beyond the range of their associated sporophytes (Nitta *et al.,* 2021), exhibiting physiological niche space larger than the sporophyte (Nitta *et al.,* 2021; Watkins *et al.,* 2007; Watkins, Mack, and Mulkey, 2007), and playing important roles in mico-scale community ecology (Harrington *et al.* 2022).

Our limited knowledge of *in situ* gametophyte biology is due in large part to the difficulty of finding and identifying them in the field. These difficulties are mainly a result of their small size and phenology. While some thalli can live for decades in tropical settings, others are short-lived and provide a narrow window during which they can be found (Farrar *et al.,* 2008; Watkins and Cardelús 2012). It can be difficult to assign unknown fern gametophytes to genus based on morphology alone since some taxa have no obvious distinguishing characters (*e.g.* Wakins and Moran, 2011). Such limited morphological differentiation and the occurrence of similar characters in distantly related taxa often preclude unequivocal identifications. The absence of dichotomous keys and a poor understanding of morphological plasticity within and between taxa add to the challenge of identifying field-collected gametophytes. Approaches using DNA sequencing to determine taxonomic identity may provide a solution (Schneider and Schuettpelz, 2006; Li *et al.,* 2009; Nitta and Chambers, 2022), but are not always possible and in the near-term will not solve the issue of rapid field identification. Broader field observations on gametophytes are needed that can later be combined with molecular studies for proper identification.

Fortunately, in some cases, gametophyte morphology can be linked to both phylogeny and ecology. For example, gross gametophyte morphology is known for representatives of more than 90% of fern genera (Nayar and Kaur, 1971), and we know that gametophyte morphology has strong correlations

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TABLE 1. Summary of morphological features observed for the six species of fern gametophytes. Farrar Type derived from Farrar *et al.* 2008.

Species	Farrar Type	Shape	Meristem	Gemmae (+/–)	Hairs (+/– and Type)	Hair Location (Surface, Margin)
Elaphoglossum latifolium	II	Strap	Notch	Absent	Capitate Glandular	Surface and Margin
Hymenophyllum myriocarpum	III	Ribbon	Marginal	Absent	Unicellular	Marginal
Lomariopsis maxonii	VI	Strap	Notch	Absent	Absent	Absent
Polypodium dulce	VI	Elongate Cordiform	Notch	Absent	Unicellular	Marginal
Polytaenium cajenense	V	Gemmiferous Ribbon	Marginal	Present	Absent	Absent
Sticherus bifidus	VI	Elongate Cordiform	Notch	Absent	Absent	Absent

with phylogeny (see Farrar *et al.* 2008). Most of this work is derived from studies of lab-grown material and we lack a more fine-scale resolution below the generic-level in most lineages. These limitations prevent more large-scale phylogenetic analysis of fern gametophyte morphology and, by extension, a complete view of fern evolution. Understanding the range and variation of morphology can aid beyond identification and phylogenetics. For example, gametophytes of epiphytic ferns, commonly exposed to periods of drought, have evolved a three-dimensional morphology (Watkins *et al.*, 2007; Pinson *et al.,* 2017) with complex branching, overlapping wings, proliferations, and trichomes, which influences water retention and perhaps results in greater tolerance to desiccation (Watkins *et al.*, 2007; Hietz, 2010; Pitterman et al., 2013). Such studies suggest that there are important form and function relationships in fern gametophytes that warrant additional study and further demonstrate the need to better understand how to identify material in the field.

The goal of this paper is to add to the growing body of work on *in situ* gametophyte biology. We aim to demonstrate ways to identify field collected gametophytes and describe *in situ* gametophyte morphology. We do this for several taxa from an array of tropical forest types in Costa Rica in an effort to expand the knowledge base on fern gametophyte morphology.

MATERIALS AND METHODS

Gametophytes of all species in this study were collected *in situ* at various sites during the 2022 Organization for Tropical Studies (OTS) Tropical Ferns and Lycophytes course. In all cases populations of many dozens to hundreds of gametophytes were found in the field and a series of careful observations

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TABLE 1. Extended.

Rhizoids (Origin Margin vs Cushion, Color)	Morphology 3D-Shape	Antheridia Morphology	Observed Sex(es)	Collection Location
Cushion, auburn	Planar	Stalked	Archegoniate, Antheridiate	Las Cruces Biological Station
Marginal, tan	Planar	Sessile	Archegoniate, Antheridiate, Hermaphroditic	Las Alturas Biological Station
Marginal, tan	Planar	Sessile	Antheridiate	Las Alturas Biological Station
Surface and marignal, Auburn	Planar	Sessile	Archegoniate, Hermaphroditic	Las Cruces Biological Station
Marginal, tan	Planar	Not observed	Sterile	Cerro Paraguas
Surface and marignal, Auburn	Trough-like	Sessile	Archegoniate, Antheridiate, Hermaphroditic	Cerro Paraguas

of these populations were carried out to trace young attached sporophytes of increasing size and maturity. This developmental series was followed through to more mature sporophytes that could be identified to species. Here, we describe only taxa for which we confidently identified using this technique, although we observed many more species during the course. With ample observed material, we are confident that we have indeed collected gametophytes of these taxa. This technique has been highly effective in identifying and describing previously unknown gametophytes (see for example: Watkins, Testo, and Merkhofer, 2011; Testo and Sundue 2014; Watkins and Moran 2019).

Gametophytes were typically collected within 0.5m of the sporophyte of the same species, but often much closer and in the immediate vicinity. Gametophyte populations were photographed in the field, collected, and a small subset brought back to the lab where we made a series of wet mounts and examined the samples to identify morphological traits. We observed these gametophytes and photographed them using compound and stereoscopic microscopes. Most images were taken using cellular phone cameras by either mounting the phone to the microscope with a general phone-ocular mount or by simply manually holding the camera to the ocular until the image came into focus. The collection sites and taxa collected for study are (Table 1): *Elaphoglossum latifolium* (Sw.) J.Sm., and *Polypodium dulce* Poir., from Las Cruces Biological Station, San Vito, Costa Rica; *Polytaenium cajenense* (Desv.) Benedict (hereafter, *Pp. dulce, Pt. cajenense*), and *Sticherus bifidus* (Willd.) Ching from Cerro Paraguas, near Las Cruces Biological Station, San Vito, Costa Rica; *Hymenophyllum myriocarpum* Hook and *Lomariopsis maxonii* (Underw.) Holttum at Las Alturas Biological Station, Costa Rica. Unmounted vouchers of gametophytes and sporophytes of all species were deposited at the L.D. Gomez Herbarium at Las Cruces.

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Results

Elaphoglossum latifolium.—Dozens of wild gametophytes from a single population growing epiphytically were collected at around 2000 m.a.s.l. from the base of an angiosperm tree in Las Alturas Biological Station, Puntarenas Province, Costa Rica (Fig. 2A–B). The gametophytes were strap-shaped and strongly elongated with a cordiform apex (Fig. 1A; Fig. 2C–F; Table 1). Gametophytes lacked a prominent midrib and were not observed to branch. The cordiform apex was partially lifted while the distal section stayed below the vegetation or flattened on the soil surface, firmly attached by exceptionally long rhizoids (Fig. 1A, Fig. 2F). The rhizoids emerged from the ventral surface and had a brown-orange (auburn) coloration. The margin and surface had capitate, secretory glandular hairs (Fig. 1C; Fig. 2H). The observed gametophytes were all unisexual, with either antheridia (Fig. 2G) or archegonia (Fig. 2I) located on the ventral surface. We discovered the occurrence of unusual stalked antheridia on male thalli (Fig.1B; Fig. 2G). These stalked antheridia were connected to the thallus by a narrow basal cell. This cell broaded into a cone shape at its distal end, and terminated with a single antheridium topped with a single cap cell. While we are aware of considerable morphological variation in antheridia in general, we believe this represents the first report of this tapered stalked form in the genus and perhaps ferns in general.

Hymenophyllum myriocarpum.—Several dozen gametophytes and sporophytes (Fig. 3A) of *H. myriocarpum* were found growing epiphytically in the understory on an unidentified angiosperm trunk along a high-light trailside. The gametophytes of this species were ribbon-shaped, produced several adventitious lateral branches (Fig. 1D, Fig. 3BG–F; Table 1), and had long brown marginal rhizoids (Fig. 1E; Fig. 3G–I). Mature gametophytes were branched and frequently overlapping (Fig. 3F). While most gametophytes were asexual, we discovered unisexual antheridiate (Fig. 3H) and archegoniate thalli (Fig. I–J), and only one hermaphroditic gametophyte. Antheridia were widespread over the surface of the thallus, although several were also found on the margins (Fig. 3H). When encountered, archegonia were always marginal and produced on thickened cushions (Fig. 3I–J). Interestingly, on bisexual thalli, antheridia and archegonia did not grow inter-mixed; we consistently found them separated by several millimeters.

Lomariopsis maxonii.—Pieces of bark and twigs covered in gametophytes of *Lomariopsis maxonii* were found on tree trunks near sporophytes (Fig 4A). Gametophytes were abundant at this site and grew vigorously on the bark substrate (Fig. 4B), often layering on top of one another. Over 40 individual gametophytes were examined, but none were found with archegonia. In general, gametophytes were strap-shaped (Fig. 4C–D; Table 1) or shelf-like (Fig. 4B), with part of the prothallus flattened against the surface and the rest of the tissue extended away from the plane of the substrate. All gametophytes examined were either asexual or antheridiate (Fig. 4F–H). Gametophytes were glabrous and, in many cases, more rhizoids emerged from one side compared to the other (Fig. 1F). Rhizoids emerged from the margins and were apically

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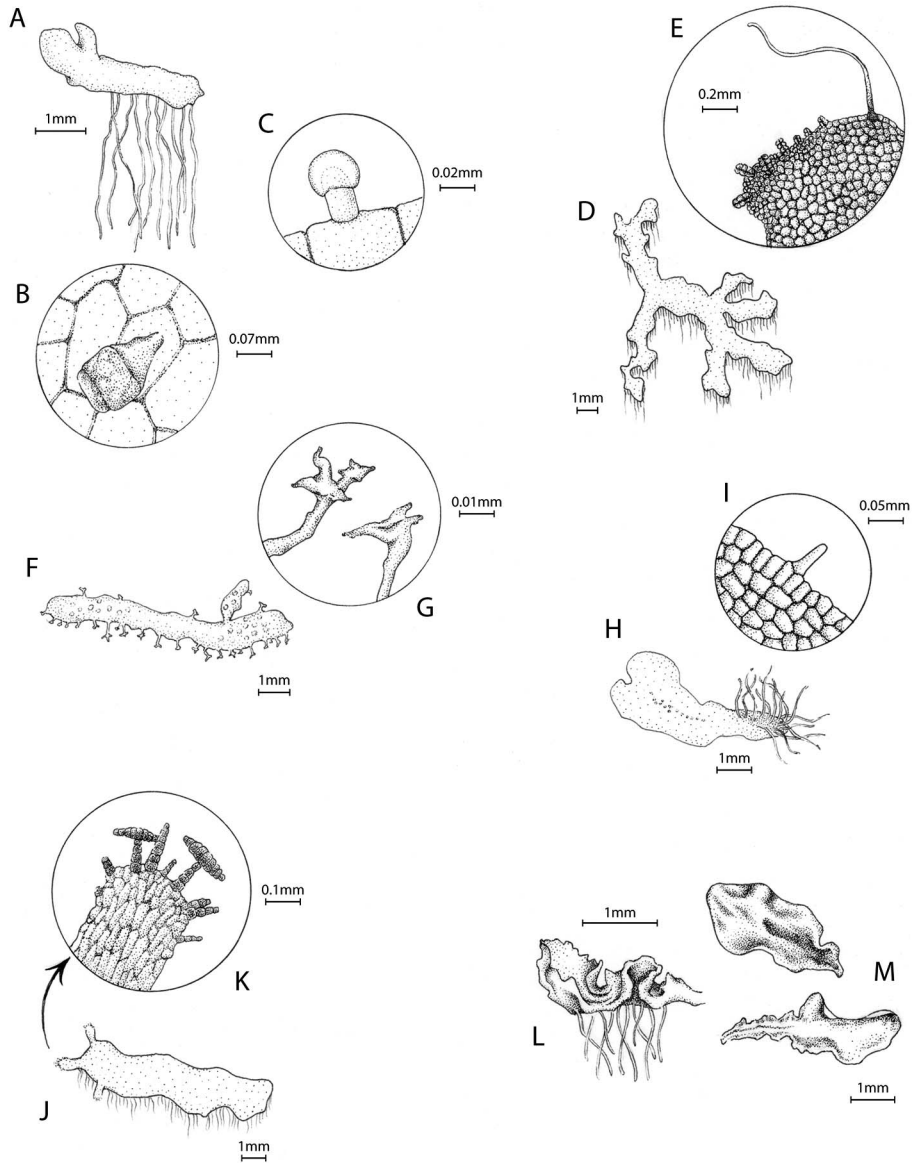


FIG. 1. Select illustrations of each gametophyte species described in this study depicting notable characters. A) *Elaphoglossum latifolium* gametophyte habit with B) stalked antheridium and C) marginal glandular hair. D) *Hymenophyllum myriocarpum* habit with E) marginal archegonia on thallus cushion and marginal rhizoid. F) *Lomariopsis maxonii* habit with G) apically branched rhizoids. H) *Polypodium dulce* habit with I) marginal hair. J) *Polytaenium cajenense* habit with K) gemmae on margins of lobed extensions from the thallus depicting "T-shaped" marginal gemmae. L) *Sticherus bifidus* habit from the side demonstrating the folder nature of the thallus and M) above, highlighting unusual 3-dimensional structure.

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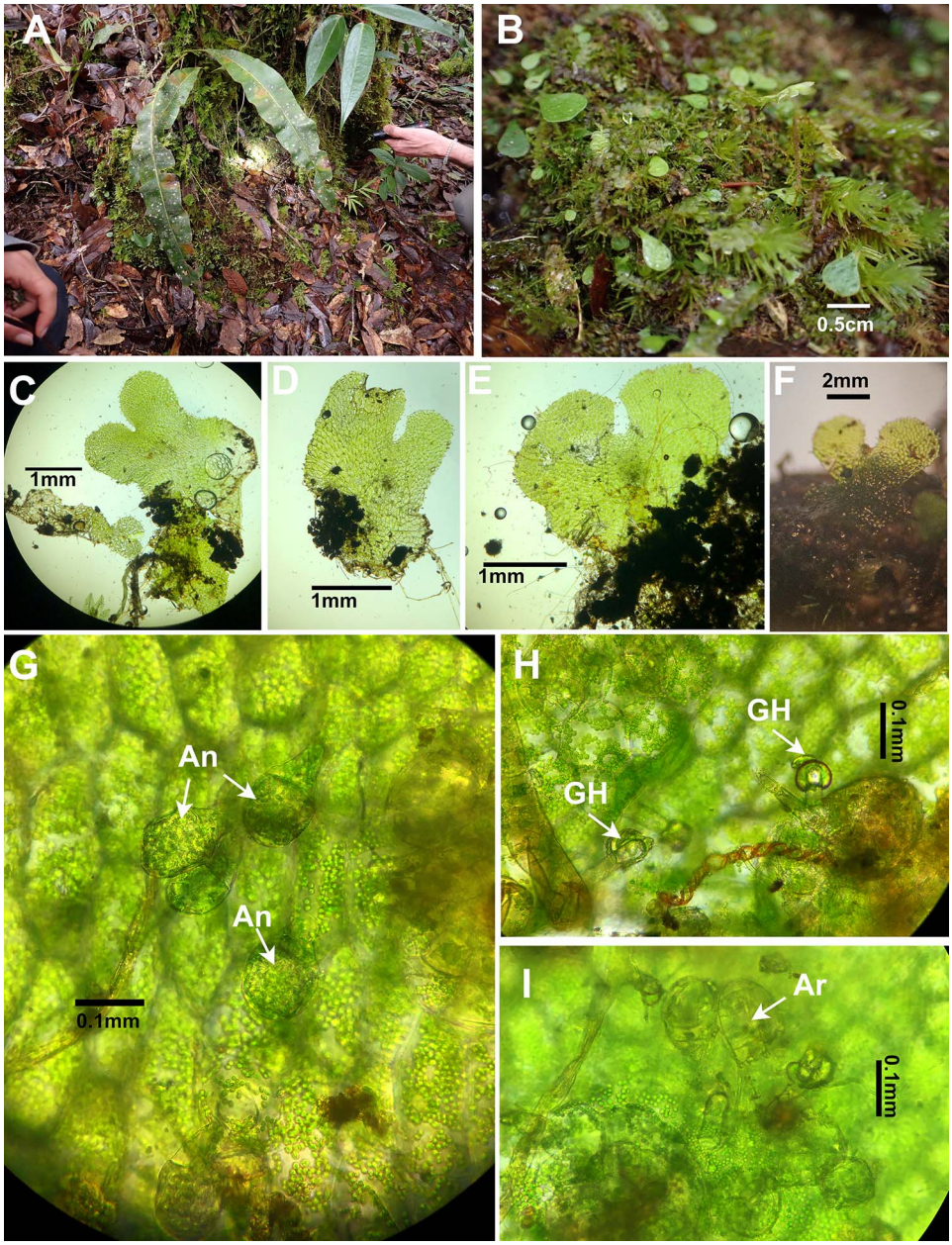


FIG. 2. *Elaphoglossum latifolium*. A) mature epiphytic sporophyte in the field. B) juvenile sporophytes, many connected to gametophytes. C–F) Series of short-elongated cordiform gametophytes. G) Collection of stalked antheridia on male thalli. H) glandular hairs common to all gametophytes and found on both margins and surface. I) Archegonia on the female thalli. An - Antheridia, GH - Glandular hairs, Ar - Archegonia.

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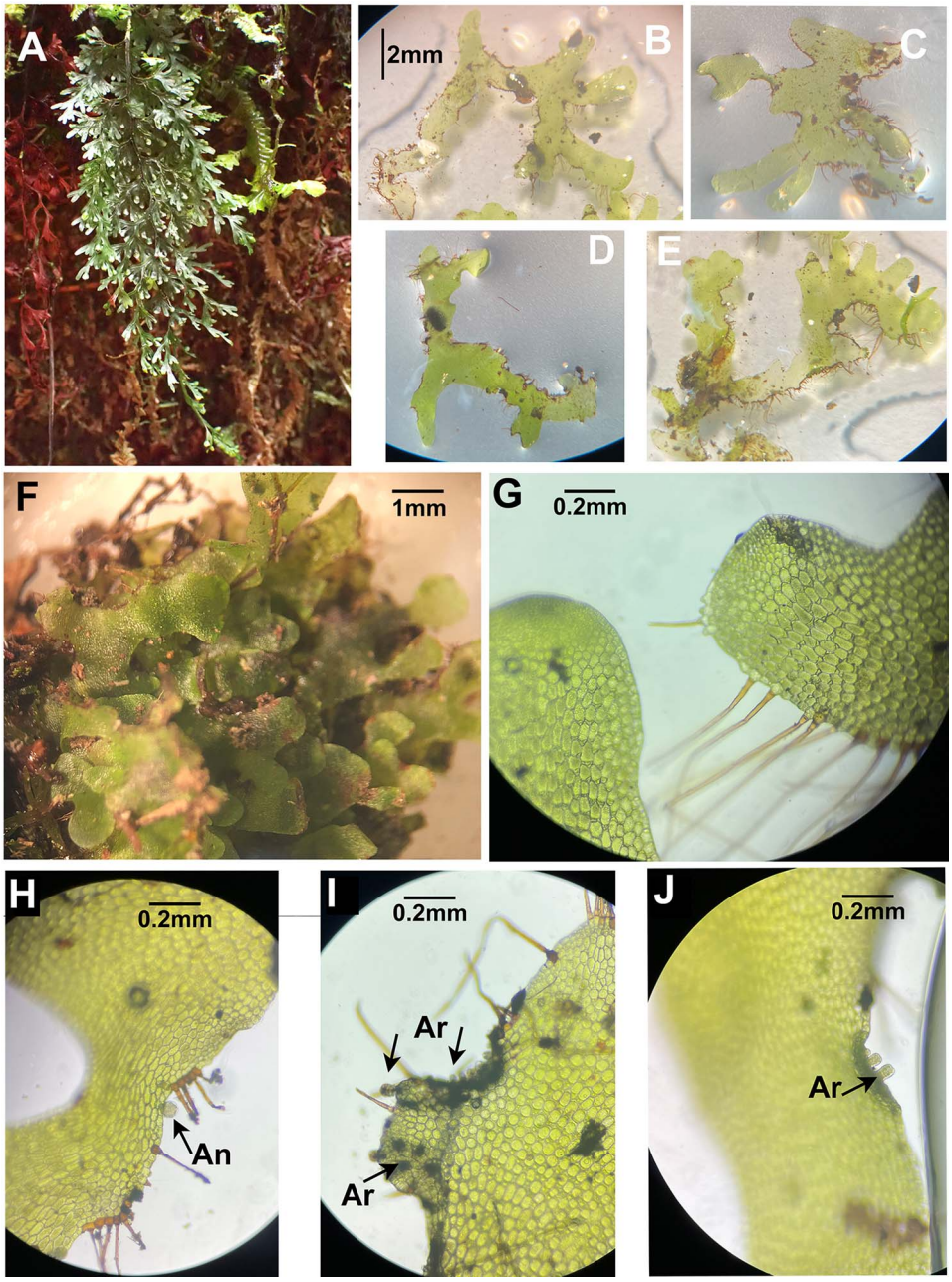


FIG. 3. *Hymenophyllum myriocarpum* A) Epiphytic sporophyte in the field. B–E) General ribbon-shape and branching gametophyte habit. F) Gametophytes were commonly found growing in clusters. G) Typical example of marginal rhizoids. H) Example of marginal antheridum and additional brown rhizoids. I–J) Marginal archegonia were born on multicellular cushions on the margins of the thallus. An - Antheridia, Ar - Archegonia.

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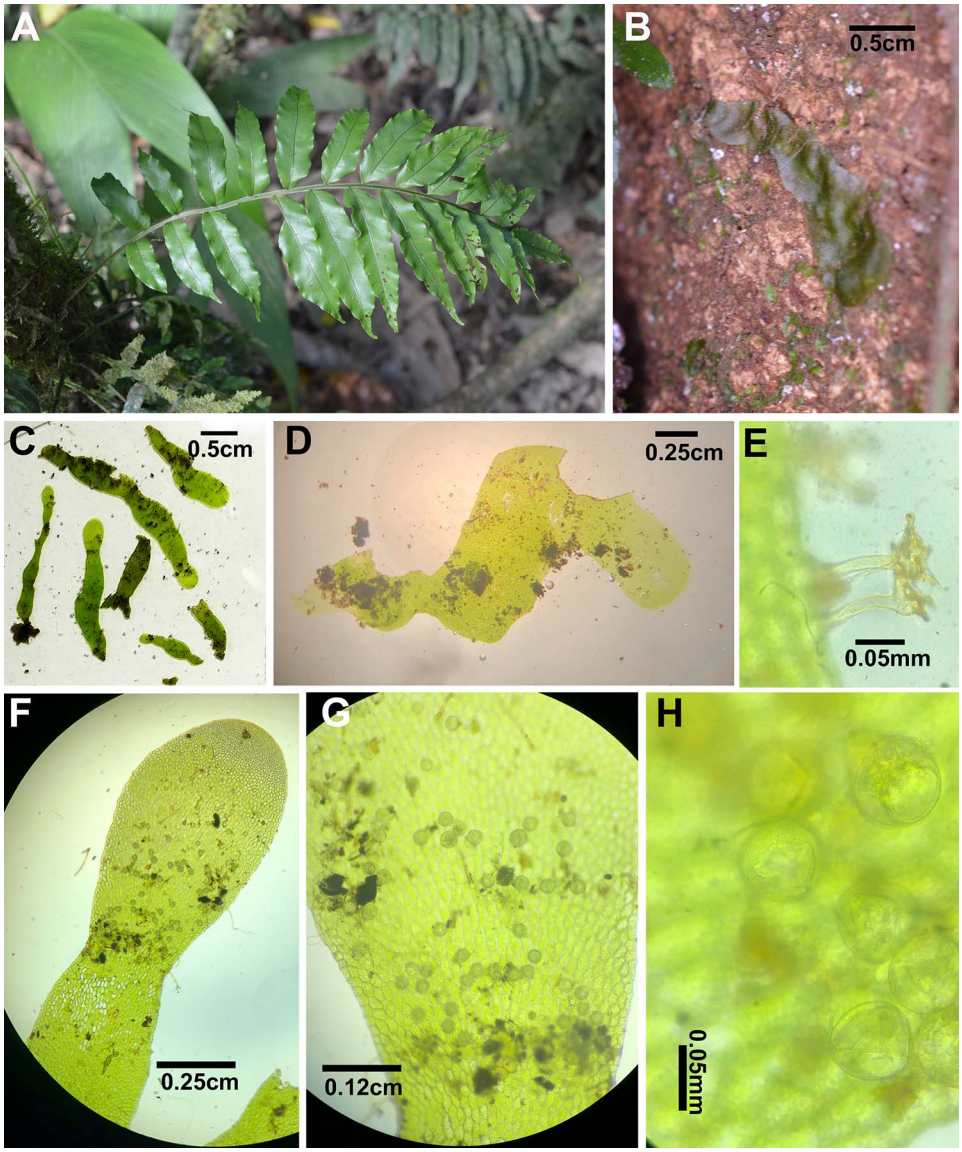


FIG. 4. *Lomariopsis maxonii*. A) Mature sporophyte growing epiphytically. B) Gametophyte with shelf-like morphology growing epiphytically. C–D) Extracted gametophytes demonstrating the strap-like morphology common in the genus. E) Apically branched rhizoids. F–H) Antheridia are conspicuous on the surface of the gametophyte.

branched (Figs. 1G, 4E). We did not observe gemmae on this species, but we did observe what appeared to be marginal branches on some established gametophytes.

Polypodium dulce.—Sporophytes of *Polypodium dulce* were discovered growing epipetrically intermixed with gametophyte populations (Fig. 5A).

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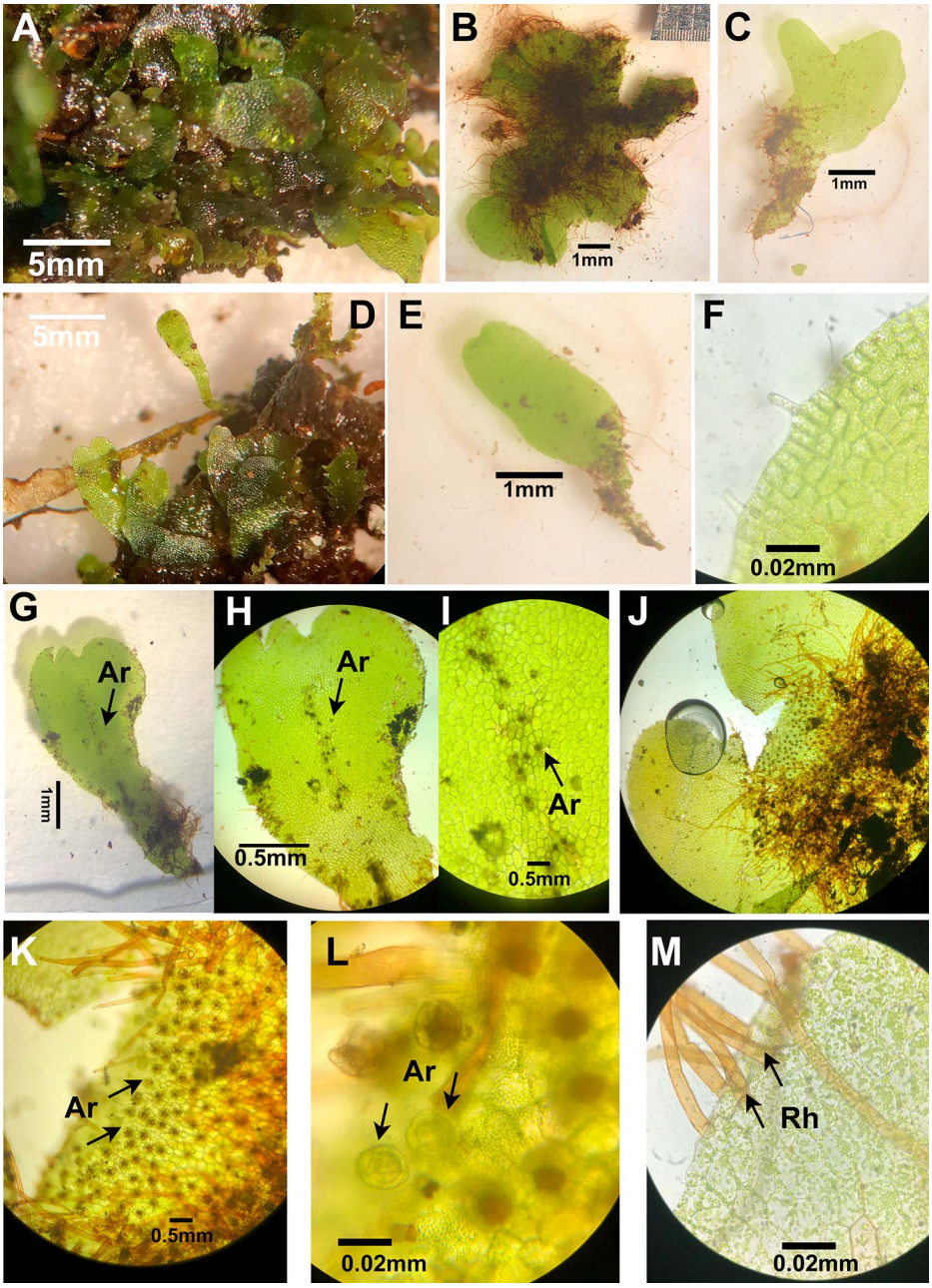


FIG. 5. *Polypodium dulce.* A–E) General elongated cordiform and strap-shaped morphology of gametophytes. F) Marginal, unicellular hair. G–I) Archegonia along midrib of elongated gametophyte. J–L) Archegonia near the apical notch of a cordiform gametophyte. L) Mix of antheridia and archegonia in hermaphroditic gametophyte. M) Auburn-colored rhizoids originating from the margin of gametophyte. Ar - Archegonia, An - Antheridia, Rh - Rhizoids.

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Gametophytes of *Pp. dulce* produce an elongated cordiform and often branching thallus (Figs. 1H; 5A–E; Table 1). Rhizoids were generally restricted to the distal portion of the gametophyte, although larger gametophytes had rhizoids originating close to the apical notch (Fig. 5B, J) or, less frequently, on the margins (Fig. 5M). In all cases, rhizoids were orange or auburn in color (Fig. 5M); all rhizoids were unbranched (Fig. 5M). The gametophytes of *Pp. dulce* had unicellular marginal hairs along the thallus surface and margins (Figs. 1I, 5F). We did not identify unisexual antheridiate gametophytes, but several archegoniate gametophytes and one hermaphroditic gametophyte were observed (Fig. 5G–L). Archegonia were located along the midrib (Figs. 1H, 5G–I) or near the apical notch (Fig. 5J,K) and could become quite abundant (Fig. 5J, K). In the hermaphroditic gametophyte, the antheridia were intermixed with archegonia near the apical notch (Fig. 5L).

Polytaenium cajenense.—We identified sporophytes and gametophytes of *Polytaenium cajenense* growing epiphytically (Figs. 6A–B) in Coto Brus, Puntarenas Province, Costa Rica. Small patches of gametophytes grew mixed with liverworts and other bryophytes (Fig. 6A–D). Gametophytes had ribbon-shaped morphology with a glabrous surface and were fixed to the substrate on one side while the other was lifted from the surface (Figs. 1J, 6B–E; Table 1). We did not find gametangia on these gametophytes. Rather, like many other vittarioid ferns (*e.g.*, Farrar, 1974), extensions of the lifted thallus produced gemmae (asexual propagules). Gemmae were T-shaped (or paired, *sensu* Crane, 1997; Fig. 6D–G) and about one millimeter in length (Figs. 1J–K; 6G–H). Each gemma contained five to six cells and two rhizoid primordial cells (i.e., rhizoid initials) (Fig. 6H). We observed a few dozen gametophytes of varying size and all thalli were sterile, thus we were unable to describe gametangia in this species.

Sticherus bifidus.—Several thousand gametophytes and an array of sporophytes of varying maturity of *S. bifidus* were found growing along a roadcut next to a large cattle pasture (Fig. 7A–C). Along the roadcut, *S. bifidus* was the only species of Gleicheniaceae with sporophytes growing in the same area. The substrate was composed of moist red soil with gametophytes growing *en masse*, frequently intertwined and overlapping in dense moss mats (Fig. 7B). Gametophytes had an elongate cordiform shape with deeply rugose/crenate margins that were up-turned away from the substrate creating a trough-like morphology (Figs. 1L–M; 7C–D, F; Table 1). Thalli were glabrous with dark brown mid-section where tough rhizoids emerged (Figs. 7D, F, G). The majority of gametophytes observed were antheridiate (Fig. 7D–E), but we did observe a few rare archegoniate thalli (Fig. 7F–H) and, rarer still, hermaphroditic thalli. Antheridiate gametophytes were short, averaging 0.5 mm, with undulating edges, markedly irregular margins, and the growth apex pointing upwards (Fig. 7B, D–E). The antheridia were located in the ventral surface (Fig. 7D), distal from the meristem and towards the edges (Fig. 7E). Archegoniate gametophytes were elongate (Fig. 7F) with undulating margins that were more irregular than the male thalli. The archegonia were located in the ventral surface (Fig. 7G), near the apical meristem and frequently on the edges (Fig. 7H).

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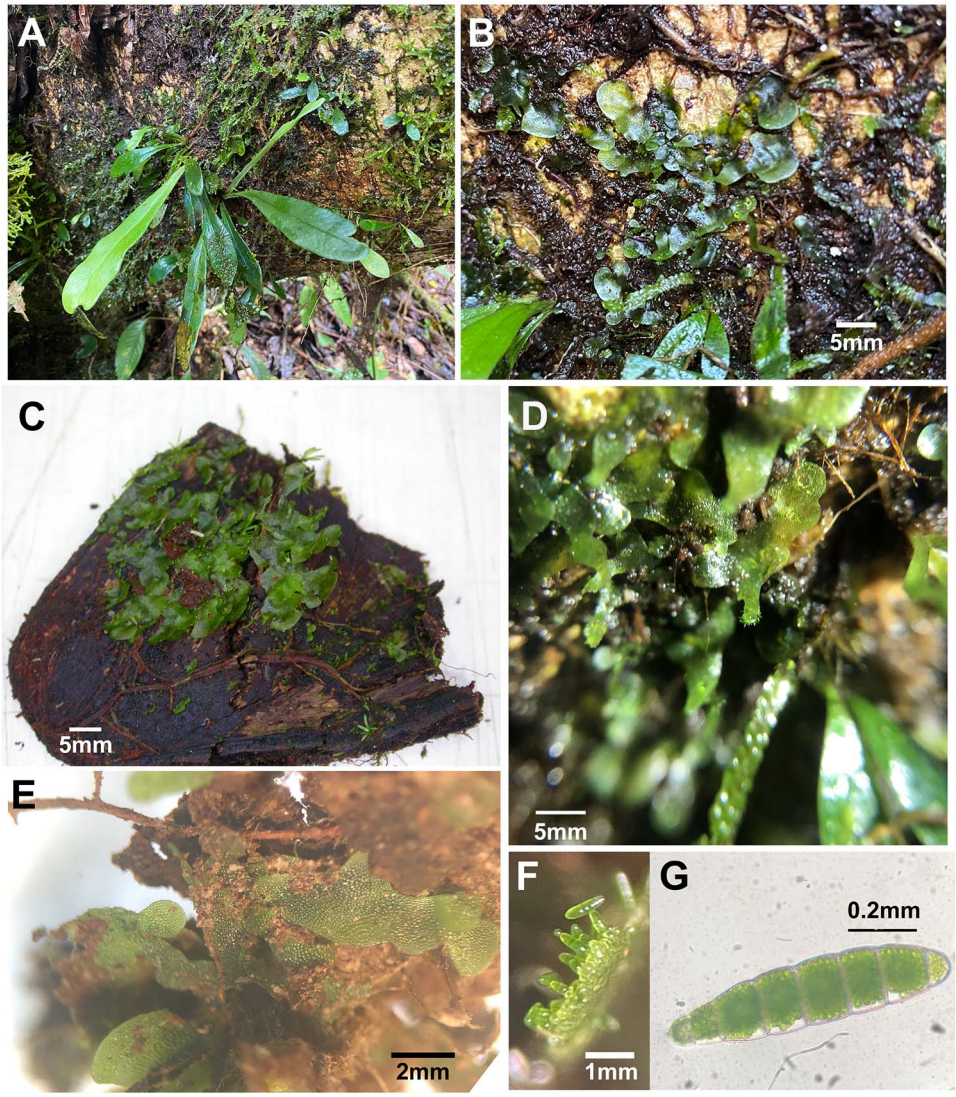


FIG. 6. *Polytaenium cajenense* A) Mature sporophyte growing epiphytically. B) Populations of *in situ* gametophytes growing epiphytically. C) Gametophytes growing along branch; note finger-like extensions from the thallus containing gemmae. D) 10X magnification of gemmae on gametophytes in the field. E) General or ribbon-shape morphology of gametophytes. F) Close-up of gemmae on gametophyte. G) Single gemma isolated from gametophyte.

DISCUSSION

Gametophyte morphology is often dismissed as unimportant due to the perception that gametophyte characters are limited, plastic, and largely uninformative for systematics. This view was first espoused by Bower (1923) and has persisted in the intervening decades, as evidenced by the relative lack of

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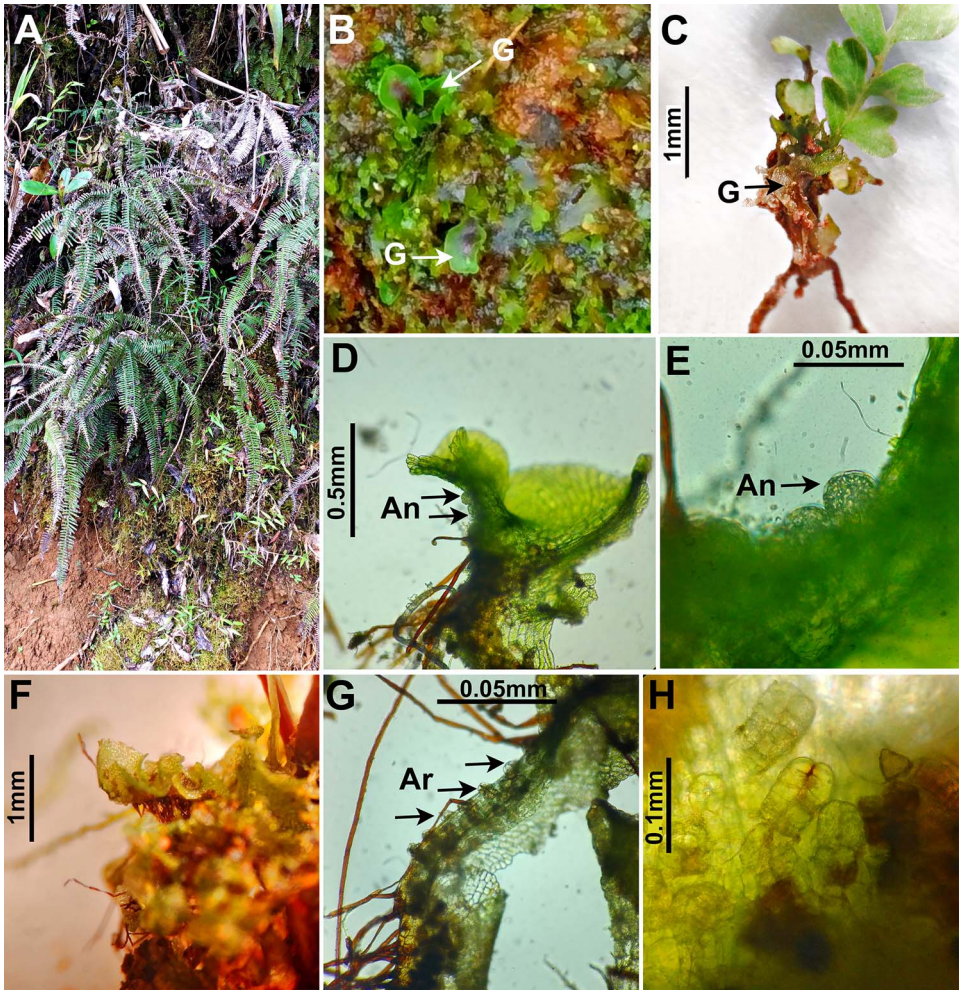


FIG. 7. *Sticherus bifidus* A) Mature terrestrial sporophytes in the field. B) Example of gametophyte cluster, forming a mass *in situ*. C) Young sporophyte with gametophyte still attached. D) Mature male gametophyte and location of the antheridia. E) Mature antheridia, with detectable spermatozoids inside. F) Mature female gametophyte demonstrating the complex folded 3D morphology commonly observed at this field site. G–H) Example of archegonia along the mid-rib of the thallus. G- Gametophyte, An- Antheridia, Ar- Archegonia.

attention fern gametophytes receive compared to sporophytes in the literature. However, our current study adds to a growing body of work demonstrating the relative consistency of gametophyte characters and the ability to use morphology to properly identify field-collected species. Moreover, our observations offer new morphological descriptions of gametophytes *in situ*, whereas most of the existing literature is largely based on morphological traits identified from lab-grown gametophytes. Ultimately, we hope that our study will encourage others to complete similar work to build a database that can be used in

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field-identification of fern gametophytes. Given that our sampling was not intended to allow for broad comparisons across taxa, we simply discuss each species examined in this study below and attempt a brief review of what is known for each taxon.

Elaphoglossum latifolium.—*Elaphoglossum* is the most diverse genus in Dryopteridaceae and one of the most species-rich among all ferns (Mickel, 1995). Despite this high species diversity, the sporophyte generation often presents similar morphology across the genus, which can make species identification using sporophytes difficult. *Elaphoglossum latifolium* belongs to one of the largest species complexes in the genus, distributed from Mexico to Bolivia, and the Caribbean islands (Tryon *et al.*, 1990). Members of the group are found from low to high elevations in wet mountain forests growing on rocks and/or as epiphytes, frequently colonizing branches in the canopy and high-light understory gaps (Woods, 2017). At the La Selva Biological Station in Costa Rica, the species has been shown to tolerate variable microhabitat conditions by inhabiting different heights in the tree, from the crown (up to 24 m) and the dark and wet buttress zone, making this an interesting subject for assessing their gametophyte morphology (Watkins and Cardelus, 2009).

The gametophytes of *Elaphoglossum* are comparatively understudied but a few descriptions have been published. Despite having similar overall morphology across the genus, there are a few species-specific differences which may prove useful for identification (*e.g.*, Stokey and Atkinson, 1957; Sánchez-Montiel, Arreguín-Sánchez, and Fernández-Nava, 2008; Lagomarsino, Grusz, and Moran, 2012; Choo, Matos, and Moran, 2014). The presence of capitate glandular hairs, lack of a prominent midrib, and absence of branching in *E. latifolium* gametophytes is similar to other *Elaphoglossum* species from published descriptions of field-collected samples (Choo, Matos, and Moran, 2014). Observations of unisexual gametophytes were also expected, given the reported response to antheridiogens of this species in cultivation (Hornych *et al.,* 2021). In general, there were few characters that distinguish *E. latifolium* from other members of the genus. However, we did discover the presence of unusual stalked antheridia and we believe this represents the first report of this tapered stalked form in the genus and perhaps ferns in general. If such a trait can be shown consistently in other populations of this species, it could be quite useful in providing taxonomic limits within this complex. Further comparisons between field collections and lab-grown material will be needed to determine the consistency of stalked antheridia morphology.

Hymenophyllum myriocarpum.—The filmy fern genus *Hymenophyllum* (Hymenophyllaceae) includes around 250 species (PPG I, 2016) that are found mostly in tropical areas; however, several species are known from temperate zones (Hennequin, 2003). Recognizing and identifying species of *Hymenophyllum* can be a difficult task due to the group's relatively homogeneous sporophyte morphology (Hennequin *et al.,* 2003). To date, relatively few *Hymenophyllum* gametophytes have been described from field observations, but *in vitro* gametophytes have been characterized in many species. Within the genus, *H. myriocarpum* is a

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widespread epiphytic species distributed from Mexico to Bolivia. This species is most often found as an understory epiphyte (Mickel and Smith, 2004), but can also grow on wet rocks along low-light stream banks (J. E. Watkins, pers. obs.).

Similar morphologies to what we report have been found for other field observations of *Hymenophyllum* species. For example, *H. wrightii* Bosch is described as long-lived with branching, ribbon-like thalli (Duffy, Stensvold, and Farrar, 2015), as are *H. badium* Hook. and Grev., *H. tayloriae* Farrar and Raine, and *H. recurvum* Gaudich. (Pinson *et al.,* 2017). The brown rhizoids of *H. myriocarpum* match descriptions of *H. pulcherrimum* Col. (Holloway, 1930). However unlike *H. myriocarpum, H. wrightii* produces linear gemmae at the margins of the thallus (Lee, Lee, and Hwang, 2020). *Hymenophyllum eximium* Kunze and *H. taloriae* also produce gemmae (Pinson et al. 2017). Not all members of the genus produce gemmae and *H. myriocarpum* did not appear to produce gemmae in the samples we collected. We carefully examined thalli for gemmifers and other scars indicating the presence of fallen gemmae; however, we failed to find this. Thus, we are fairly certain that gametophytes of this taxon do not produce gemmae.

Lomariopsis maxonii.—The genus *Lomariopsis* has a wide tropical and subtropical distribution, with members extending from the American tropics, into Africa, Asia and Oceania, and islands of the Indian Ocean (Wu *et al.,* 2021). This genus is the most species-rich within the family Lomariopsidaceae, with approximately 60 species (PPG I, 2016). *Lomariopsis* is distinguished from other genera in that all species in the genus are hemiepiphytic and produce long-creeping and/or climbing rhizomes, once-pinnate leaves (with one exception, *L. palustris* Mett. ex Kuhn), a conform terminal pinna, jointed lateral pinnae, parallel free veins, and light-colored, cartilaginous margins (see Moran, 2000 for full description). Some members of the genus are known to produce gametophytes capable of persisting as long-lived asexual colonies (Watkins and Moran, 2019), with at least two species known to exist as independent gametophyte-only populations in some subtropical habitats (*e.g., L. kunzeana* (C.Presl) Holttum and *L. lineata* (C.Presl) Holttum; Pinson *et al.,* 2017).

Branched rhizoids in *L. maxonii* are very similar to other *Lomariopsis* species and are presumably an adaptation that helps gametophytes adhere to the epiphytic surface (*e.g.,* Watkins and Moran, 2019). Although *L. maxonnii* does not seem to produce gemmae, the gametophyte thallus branches in a way that seems to generate distinct lobes. Since these gametophytes are known to form long-lived colonies, it is possible that these branches may detach from the original gametophyte to form an independent individual. Similar to other *Lomariopsis* species (Watkins and Moran, 2019), *L. maxonii* gametophytes lack unique features, making it difficult to distinguish from gametophytes of similar species without observing attached sporophyte material or using another method, such as DNA barcoding (Li *et al.,* 2009; Nitta and Chambers, 2022).

Polypodium dulce.—The genus *Polypodium* belongs to one of the largest fern families, Polypodiaceae, which is composed of nearly 1700 species (PPG I, 2016). *Polypodium* is globally distributed with a center of diversity in the

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American tropics, although the temperate *Pp. vulgare* L. complex has been extensively studied as a model system for reticulate evolution in ferns (*e.g.*, Manton, 1947; Haufler, Windham, and Rabe 1995; Sigel *et al.,* 2014). There are an estimated 40 species in *Polypodium* (PPG I, 2016), most of which are epiphytic or epipetric. Members of the *Pp. dulce* group have short-creeping rhizomes with pubescent scales, 10-25 segments of the lamina, ex-indusiate sori, and yellow spores. This group contains about 20 species, all in the American tropics, is most diverse in Mexico, and is named after the most common and widespread species *Pp. dulce.* Gametophyte morphology has been described in several temperate (e.g., *Pp. vulgare;* Pickett and Thayer, 1927) and tropical species (*e.g.*, *Pp. lepidotrichum* Maxon*;* Jaramillo and Pérez-García, 1994), but the gametophytes of *Pp. dulce* have not yet been described.

The morphology of the gametophytes with an elongated cordiform shape, meristem at the apical notch, and marginal unicellular hairs is relatively consistent with other *Polypodium* species (*e.g.*, Jaramilla and Pérez-García, 1994, Chiou and Farrar 1997, Vásquez-Pérez, Mendoz-Ruiz, and Pérez-García, 2012). Although Chiou and Farrar (1997) reported branched rhizoids in some species of Polypodiaceae, the gametophytes of *Pp. dulce* have unbranched rhizoids. We only observed archegoniate and hermaphroditic gametophytes, though Chiou, Farrar, and Ranker (2002) found mixes of antheridiate, archegoniate, and hermaphroditic gametophytes of *Pp. pellucidum* Kaulf. Our observation and description of gametophytes is the first for this common, widespread species and is largely consistent with the morphology described for other *Polypodium* gametophytes.

Polytaenium cajenense.—The vittarioid ferns are a clade of about 350 mostly epiphytic species with relatively reduced morphology within Pteridaceae (PPG I, 2016; Schuettpelz *et al.*, 2016). Indeterminate growth and gemmae production are common among many vittarioid fern gametophytes and allow them to persist for years or indefinitely without producing a sporophyte (*e.g.*, Farrar 1967, Farrar *et al.*, 2008). In the genus *Polytaenium* there are about ten species distributed throughout tropical Central and South America (Schuettpelz *et al.*, 2016). *Polytaenium cajenense* is a widespread member of the genus with a range from southern Mexico and the Caribbean islands through southern Brazil which tends to occur low on tree trunks in wet forests. At least two species of *Polytaenium* form a relationship with ants (Watkins, Cardelús, and Mack, 2008; León and Young, 2010), including *Pt. cajenense.* Although this species is widely distributed, it is considered critically endangered in some areas of its range (Pereira *et al.,* 2013) and relatively little is known about its biology.

To date, the only description of *Polytaenium* gametophytes comes from Drassler (1995) on *Pt. cajenense.* Although we made *in situ* collections and Drassler (1995) made observations from a lab-grown culture, the gametophyte morphology is largely consistent. The only difference we noted was the number of cells in the gemmae; we observed five to six cells per gemma with a rhizoid initial at each end, while Drassler (1995) documented four to five cells per gemma with a rhizoid initial at one or both ends. Unlike some *Vittaria*

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species (Emigh and Farrar, 1977), gametangia were not present on gemmae in either the lab-grown (Drassler, 1995) or field-collected gametophytes.

Sticherus bifidus.—*Sticherus* is the most species-rich genus in the family Gleicheniaceae, with around 95 species and a cosmopolitan distribution (PPGI, 2016). In the Americas, *Sticherus* species are found from Mexico to Patagonia, but with a disjunct distribution in South America with approximately six isolated species in the Patagonian and Sub-Antarctic regions (Gonzales and Kessler, 2011). Species typically grow in open, disturbed environments within montane forests, and frequently prefer clearings, roadcuts, or other highly disturbed areas (Gonzáles and Kessler, 2011).

What was once the more widely recognized taxon of *Sticherus bifidus* is now divided into seven distinct species and maintains the capacity for widespread hybridization. The species is most likely a complex and this has hampered the study of *S. bifidus sensu stricto* (Gonzáles and Kessler, 2011). Thus, gametophytes may prove useful in taxon delineations in this group. Curiously, earlier work on this taxon reported the presence of multicellular hairs in lab cultures (Stokey, 1950, Farfán-Santillan *et al.,* 2017). Stokey (1950) reported that these hairs typically appeared after around two months in culture, and noted that hair production slows with age and hairs can shrivel and may be covered over by rhizoids. That same study reported that hair presence can be quite variable in some species. For example, in what Stokey terms "weaker cultures," gametophytes of *S. palmatus* were glabrous; however "stronger" cultures produced copious hairs. Stokey reported hairs in eight of the ten species examined, and speculated that the two glabrous taxa had simply not developed enough to produce hairs with the overall assumption being that members of the family all produce gametophytic hairs. In our study, all of the gametophytes examined were glabrous. The lack of hairs across what was a clear developmental series is curious and it is possible that the conditions at this site were not right for hair production. Our observations taken with those of Stokey, suggest that more work needs to be done comparing hair production across growing conditions.

Finally, an interesting feature of the gametophytes we observed is their trough-like morphology, which was previously recorded for this species and is common among the genus (Stokey, 1950, Farfán-Santillán *et al.,* 2017). We suspect that this morphology may enhance water retention at the center of the gametophyte, which could be a strategy for exohydric water-holding capacity. Such a low surface-to-volume ratio of this morphology may represent a strategy for surviving environmental fluctuations in bare soil habitats (Haufler and Adams, 1982).

Conclusion

It is our hope that this work will spur more interest in field collections of fern gametophytes. Much of what we know about morphology and gametophyte ecology is derived from *in vitro* cultures. While we did not expect many characters to vary significantly between lab and field populations, our

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observations of no hair production in field-collected *Sticherus* suggests that much more work needs to be done to assess the impact of culture conditions on aspects of trait morphology, such as hair production and density. Based on previous work, researchers studying fern gametophyte morphology might assume that hairs are a constant feature of a species that can be applied for identification. Variation of hair presence within a species is therefore an important discovery that requires further work. It is also our intention to demonstrate that it is entirely feasible to find and identify some taxa in field settings. Of course, this is most true of tropical habitats where conditions are right for dense long-lived gametophyte populations. It will always be frustratingly difficult to do such work in temperate floras and even the most self-assured pteridologist would be fool-hardy to suggest that it is easy to identify most taxa from field collections without molecular tools. However, the field would greatly benefit from more studies of the type here to understand better the extent and range of gametophyte variation in field settings.

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